

WARRENTON-FAUQUIER, VA, USA

WMO: 722692

Lat: 38.586N

Lon: 77.711W

Elev: 338

StdP: 14.52

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00367

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	17.6	-2.4	4.9	24.3	2.1	6.2	26.8	23.2	33.7	20.4	35.4	2.2	300	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	95.3	76.1	92.7	75.1	90.3	74.2	81.5	90.3	79.7	86.7	77.7	84.6	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	153.6	86.1	77.4	144.4	83.7	75.9	136.9	82.0	45.5	92.1	43.4	86.6	41.3	83.9	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.1	12.9	4.9	98.8	6.4	3.4	0.3	101.2	-3.5	103.2	-7.0	105.1	-11.7	107.6		
(5)			4.2	81.3	6.4	3.7	-0.4	84.0	-4.1	86.2	-7.7	88.2	-12.3	90.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.0	34.2	39.6	45.6	57.0	66.2	73.9	78.1	75.8	69.7	57.7	45.7	39.3	(6)
(7)		DBStd	17.08	9.69	9.80	10.11	8.92	7.83	5.37	5.12	4.84	6.73	8.43	8.27	8.98	(7)
(8)		HDD50	1609	499	317	210	30	1	0	0	0	0	22	177	353	(8)
(9)		HDD65	4303	955	713	607	274	84	4	0	0	31	259	579	797	(9)
(10)		CDD50	4158	9	25	72	239	504	718	870	798	590	262	49	22	(10)
(11)		CDD65	1376	0	1	4	32	122	272	405	334	171	34	1	0	(11)
(12)		CDH74	14060	0	9	63	447	1233	2706	4339	3295	1614	342	11	1	(12)
(13)		CDH80	5950	0	0	12	137	421	1100	2100	1435	648	97	0	0	(13)
(14)	Wind	WSAvg	4.3	5.0	5.4	6.1	5.5	4.2	3.6	2.9	2.9	3.4	4.0	4.2	4.1	(14)
(15)	Precipitation	PrecAvg	41.3	3.0	2.2	3.8	3.3	4.1	4.0	4.1	3.3	4.1	3.1	3.2	3.2	(15)
(16)		PrecMax	56.4	6.6	6.5	9.0	5.0	8.2	8.4	7.1	7.3	10.2	8.1	6.2	5.7	(16)
(17)		PrecMin	29.4	1.0	0.6	0.0	1.6	1.4	1.4	1.3	1.2	0.0	0.1	0.9	1.6	(17)
(18)		PrecStd	6.2	1.6	1.2	2.0	1.1	1.9	1.8	1.6	1.4	2.6	2.0	1.6	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.1	75.0	81.1	89.7	91.2	96.9	100.2	96.8	94.9	87.5	75.1	70.4	(19)
(20)			MCWB	58.9	59.6	66.3	66.3	73.9	79.6	81.6	76.2	72.0	71.4	62.1	65.5	(20)
(21)		2%	DB	61.6	67.9	73.5	84.1	88.2	92.2	96.9	93.1	91.3	82.3	70.0	64.7	(21)
(22)			MCWB	54.8	56.3	59.4	64.8	73.3	75.9	77.7	74.1	72.7	68.7	59.5	60.9	(22)
(23)		5%	DB	55.4	62.2	69.7	79.3	85.1	89.8	93.4	90.6	87.8	77.3	65.3	60.3	(23)
(24)			MCWB	48.3	52.3	58.3	63.4	71.3	73.8	76.7	73.8	71.0	64.6	56.9	55.0	(24)
(25)		10%	DB	49.9	55.4	64.0	74.3	81.5	86.6	90.5	88.3	83.2	73.1	61.3	54.7	(25)
(26)			MCWB	44.6	47.8	54.6	60.7	69.4	72.4	75.3	73.3	68.8	64.2	53.8	49.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.3	61.9	69.7	73.6	80.3	82.3	84.8	81.5	80.7	74.4	66.3	66.9	(27)
(28)			MCD B	65.8	69.3	75.2	82.9	89.0	93.1	96.6	90.1	89.0	82.8	69.9	68.9	(28)
(29)		2%	WB	55.4	58.5	63.8	69.8	76.4	79.8	82.7	79.3	77.4	71.8	62.8	63.1	(29)
(30)			MCD B	60.7	66.5	68.6	78.6	83.5	88.0	92.3	85.1	83.3	78.7	67.1	65.3	(30)
(31)		5%	WB	49.2	53.1	59.7	66.0	73.7	77.5	80.7	77.5	74.7	69.3	59.1	56.2	(31)
(32)			MCD B	54.6	60.5	68.3	74.7	81.3	85.1	88.3	84.6	81.9	74.6	63.7	59.2	(32)
(33)		10%	WB	45.0	48.7	55.2	62.7	71.5	75.1	78.3	75.7	72.6	66.1	54.5	50.5	(33)
(34)			MCD B	49.6	55.4	62.7	71.7	78.7	82.8	84.8	83.8	78.6	71.9	59.5	53.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.6	20.8	23.0	25.2	22.3	22.2	22.1	22.5	22.7	24.3	23.0	19.7	(35)
(36)			MCDBR	28.0	29.5	30.8	31.2	26.4	25.2	25.8	25.4	28.8	28.9	30.1	24.8	(36)
(37)			MCWBR	20.7	19.6	18.2	15.0	13.2	11.4	9.9	9.6	12.2	15.7	19.5	19.7	(37)
(38)		5% WB	MCD BR	23.5	25.5	26.0	25.5	22.8	22.4	21.9	20.3	21.2	22.9	22.9	22.1	(38)
(39)			MCWBR	19.1	18.5	18.5	14.5	12.7	11.8	10.5	9.7	11.1	14.5	17.6	19.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.317	0.352	0.393	0.432	0.490	0.505	0.507	0.418	0.356	0.319	0.305	(40)	
(41)		taud	2.498	2.432	2.392	2.302	2.218	2.096	2.076	2.052	2.281	2.454	2.513	2.520	(41)	
(42)		Ebn at Noon	280	289	288	281	271	254	249	245	262	271	270	270	(42)	
(43)		Edh at Noon	25	30	35	40	45	51	51	51	38	29	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	675	935	1244	1554	1748	1925	1872	1678	1379	1048	765	594	(44)	
(45)		RadStd	59	104	82	116	176	141	100	82	166	118	62	56	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+130

Nomenclature: See separate page